MA1522 Linear Algebra for Computing
L2 Lesson 14



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Semester 1, 2026/27

Coordinates Relative to an Orthogonal Basis

Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$ be an orthogonal subset of $\mathbb{R}^n$, and suppose that $\boldsymbol{u}_i \neq 0$ for all i.

Then S is **linearly independent** and is therefore a **basis** for $\operatorname{span}(S)$.

Let $\boldsymbol{v} \in \mathbb{R}^n$. Then $\boldsymbol{v} \in \operatorname{span}(S)$ if and only if

$$\boldsymbol{v} = \frac{\boldsymbol{v} \cdot \boldsymbol{u}_1}{\|\boldsymbol{u}_1\|^2}\,\boldsymbol{u}_1 + \frac{\boldsymbol{v} \cdot \boldsymbol{u}_2}{\|\boldsymbol{u}_2\|^2}\,\boldsymbol{u}_2 + \cdots + \frac{\boldsymbol{v} \cdot \boldsymbol{u}_k}{\|\boldsymbol{u}_k\|^2}\,\boldsymbol{u}_k,$$

in which case $[\boldsymbol{v}]_S = \begin{pmatrix} \frac{\boldsymbol{v} \cdot \boldsymbol{u}_1}{\|\boldsymbol{u}_1\|^2} \\ \frac{\boldsymbol{v} \cdot \boldsymbol{u}_2}{\|\boldsymbol{u}_2\|^2} \\ \vdots \\ \frac{\boldsymbol{v} \cdot \boldsymbol{u}_k}{\|\boldsymbol{u}_k\|^2} \end{pmatrix}$.

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Question 1

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Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4\}$, where

$$\boldsymbol{v}_1 = (1, 2, 2, -1),\ \boldsymbol{v}_2 = (1, 1, -1, 1),\ \boldsymbol{v}_3 = (-1, 1, -1, -1),\ \boldsymbol{v}_4 = (-2, 1, 1, 2).$$

a. Show that S is an orthogonal basis for $\mathbb{R}^4$.

[Pg 108 of Chapter 3, Pg 24 of Chapter 5]

b. Find the coordinate vector of $\boldsymbol{v} = (0, 1, 2, 3)$ relative to S.

[Pg 27–30 of Chapter 5]

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Solution.

a. We check that $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = \boldsymbol{v}_1 \cdot \boldsymbol{v}_3 = \boldsymbol{v}_1 \cdot \boldsymbol{v}_4 = \boldsymbol{v}_2 \cdot \boldsymbol{v}_3 = \boldsymbol{v}_2 \cdot \boldsymbol{v}_4 = \boldsymbol{v}_3 \cdot \boldsymbol{v}_4 = 0$. So S is orthogonal and hence linearly independent.

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Since $S \subseteq \mathbb{R}^4$ with $\dim(\mathbb{R}^4) = 4 = |S|$, S is a basis for $\mathbb{R}^4$.

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Since $S \subseteq \mathbb{R}^4$ with $\dim(\mathbb{R}^4) = 4 = |S|$, S is a basis for $\mathbb{R}^4$.

b. $[\boldsymbol{v}]_S = \left(\frac{\boldsymbol{v}\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}, \frac{\boldsymbol{v}\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}, \frac{\boldsymbol{v}\cdot\boldsymbol{v}_3}{\|\boldsymbol{v}_3\|^2}, \frac{\boldsymbol{v}\cdot\boldsymbol{v}_4}{\|\boldsymbol{v}_4\|^2}\right)^T = \left(\frac{3}{10}, \frac{1}{2}, -1, \frac{9}{10}\right)^T$.

Orthogonal Projection onto a Subspace

Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$ be an orthogonal subset of $\mathbb{R}^n$, and suppose that $\boldsymbol{u}_i \neq 0$ for all i.

Let $\boldsymbol{v} \in \mathbb{R}^n$. The **orthogonal projection of $\boldsymbol{v}$ onto** $\operatorname{span}(S)$ is the vector

$$\frac{\boldsymbol{v} \cdot \boldsymbol{u}_1}{\|\boldsymbol{u}_1\|^2}\,\boldsymbol{u}_1 + \frac{\boldsymbol{v} \cdot \boldsymbol{u}_2}{\|\boldsymbol{u}_2\|^2}\,\boldsymbol{u}_2 + \cdots + \frac{\boldsymbol{v} \cdot \boldsymbol{u}_k}{\|\boldsymbol{u}_k\|^2}\,\boldsymbol{u}_k.$$

Theorem

Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$ be an orthogonal subset of $\mathbb{R}^n$, and suppose that $\boldsymbol{u}_i \neq 0$ for all i. Let $\boldsymbol{v} \in \mathbb{R}^n$ and let $\boldsymbol{v}_\mathrm{p}$ denote the orthogonal projection of $\boldsymbol{v}$ onto $\mathrm{span}(S)$.

1. $\boldsymbol{v} \in \mathrm{span}(S)$ *if and only if* $\boldsymbol{v} = \boldsymbol{v}_\mathrm{p}$.
2. $\boldsymbol{v}_\mathrm{p}$ *is the unique vector in* $\mathrm{span}(S)$ *such that:*
   - $\|\boldsymbol{v}\|^2 = \|\boldsymbol{v} - \boldsymbol{v}_\mathrm{p}\|^2 + \|\boldsymbol{v}_\mathrm{p}\|^2$;
   - $\|\boldsymbol{v} - \boldsymbol{v}_\mathrm{p}\| \leq \|\boldsymbol{v} - \boldsymbol{u}\|$ *for all* $\boldsymbol{u} \in \mathrm{span}(S)$;
   - $(\boldsymbol{v} - \boldsymbol{v}_\mathrm{p}) \cdot \boldsymbol{u} = 0$ *for all* $\boldsymbol{u} \in \mathrm{span}(S)$;
   - $(\boldsymbol{v} - \boldsymbol{v}_\mathrm{p}) \cdot \boldsymbol{u}_i = 0$ *for all* i.

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Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$ *be an orthogonal subset of* $\mathbb{R}^n$*, and suppose that* $\boldsymbol{u}_i \neq 0$ *for all* i*. Let* $\boldsymbol{v} \in \mathbb{R}^n$ *and let* $\boldsymbol{v}_\mathrm{p}$ *denote the orthogonal projection of* $\boldsymbol{v}$ *onto* $\mathrm{span}(S)$*.*

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Question 2

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Let $S = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3\}$, where

$$\boldsymbol{w}_1 = (1, 1, 1, 1, 1),\ \boldsymbol{w}_2 = (1, 2, -1, -2, 0),\ \boldsymbol{w}_3 = (1, -1, 1, -1, 0).$$

a. Show that S is orthogonal.

b. Find the projection $\boldsymbol{v}_\text{p}$ of $\boldsymbol{v} = (2, 0, 1, 1, -1)$ onto $\text{span}(S)$.
[Pg 43–44 of Chapter 5]

c. Verify that $\boldsymbol{v} - \boldsymbol{v}_\text{p}$ is orthogonal to all vectors in $\text{span}(S)$.

Question 2: Solution

a. Check that $\boldsymbol{w}_1 \cdot \boldsymbol{w}_2 = \boldsymbol{w}_1 \cdot \boldsymbol{w}_3 = \boldsymbol{w}_2 \cdot \boldsymbol{w}_3 = 0$, so S is orthogonal.

b. $\boldsymbol{v}_{\mathrm{p}} = \frac{\boldsymbol{v}\cdot\boldsymbol{w}_1}{\|\boldsymbol{w}_1\|^2}\boldsymbol{w}_1 + \frac{\boldsymbol{v}\cdot\boldsymbol{w}_2}{\|\boldsymbol{w}_2\|^2}\boldsymbol{w}_2 + \frac{\boldsymbol{v}\cdot\boldsymbol{w}_3}{\|\boldsymbol{w}_3\|^2}\boldsymbol{w}_3 = \frac{3}{5}\boldsymbol{w}_1 - \frac{1}{10}\boldsymbol{w}_2 + \frac{1}{2}\boldsymbol{w}_3$
$= (1, -\frac{1}{10}, \frac{6}{5}, \frac{3}{10}, \frac{3}{5})$.

c. $\boldsymbol{n} := \boldsymbol{v} - \boldsymbol{v}_{\mathrm{p}} = (1, \frac{1}{10}, -\frac{1}{5}, \frac{7}{10}, -\frac{8}{5})$. It is straightforward to verify that

$$\boldsymbol{n} \cdot \boldsymbol{w}_1 = \boldsymbol{n} \cdot \boldsymbol{w}_2 = \boldsymbol{n} \cdot \boldsymbol{w}_3 = 0.$$

Let $\boldsymbol{w} \in \mathrm{span}(S)$. Then $\boldsymbol{w} = c_1\boldsymbol{w}_1 + c_2\boldsymbol{w}_2 + c_3\boldsymbol{w}_3$ for some $c_1, c_2, c_3 \in \mathbb{R}$. So

$$\begin{aligned}\boldsymbol{n} \cdot \boldsymbol{w} &= \boldsymbol{n} \cdot (c_1\boldsymbol{w}_1 + c_2\boldsymbol{w}_2 + c_3\boldsymbol{w}_3) \\ &= c_1(\boldsymbol{n} \cdot \boldsymbol{w}_1) + c_2(\boldsymbol{n} \cdot \boldsymbol{w}_2) + c_3(\boldsymbol{n} \cdot \boldsymbol{w}_3) = 0.\end{aligned}$$

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Gram-Schmidt Process/Algorithm

Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\} \subseteq \mathbb{R}^n$.
The **Gram-Schmidt Process/Algorithm** generates an orthogonal basis T_i for $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_i\}$ recursively, for each i, as follows:

Let $T_0 = \emptyset$, and for each $i \in \{1, \ldots, k\}$, recursively define

$$T_i = \begin{cases} T_{i-1} \cup \{\boldsymbol{u}_i - \boldsymbol{p}_i\}, & \text{if } \boldsymbol{u}_i \neq \boldsymbol{p}_i; \\ T_{i-1}, & \text{if } \boldsymbol{u}_i = \boldsymbol{p}_i, \end{cases}$$

where $\boldsymbol{p}_i$ is the orthogonal projection of $\boldsymbol{u}_i$ onto $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{i-1}\} = \operatorname{span}(T_{i-1})$.

In particular, T_k is an orthogonal basis for $\operatorname{span}(S)$.

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In particular, T_k is an orthogonal basis for $\operatorname{span}(S)$.

Gram-Schmidt in Action

Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$ be an linearly independent subset of $\mathbb{R}^n$. Let

$$\begin{aligned}
\boldsymbol{v}_1 &= \boldsymbol{u}_1, \\
\boldsymbol{v}_2 &= \boldsymbol{u}_2 - \frac{\boldsymbol{u}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\, \boldsymbol{v}_1, \\
\boldsymbol{v}_3 &= \boldsymbol{u}_3 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\, \boldsymbol{v}_1 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\, \boldsymbol{v}_2, \\
&\vdots \\
\boldsymbol{v}_k &= \boldsymbol{u}_k - \frac{\boldsymbol{u}_k \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\, \boldsymbol{v}_1 - \frac{\boldsymbol{u}_k \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\, \boldsymbol{v}_2 - \cdots - \frac{\boldsymbol{u}_k \cdot \boldsymbol{v}_{k-1}}{\|\boldsymbol{v}_{k-1}\|^2}\, \boldsymbol{v}_{k-1}.
\end{aligned}$$

Then $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\}$ is an **orthogonal** basis for $\operatorname{span}(S)$, and $\{\frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1, \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2, \ldots, \frac{1}{\|\boldsymbol{v}_k\|}\boldsymbol{v}_k\}$ is an **orthonormal** basis for $\operatorname{span}(S)$.

Question 3

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Apply the Gram-Schmidt process to convert each of the following sets into an orthogonal set in $\mathbb{R}^4$. [Pg 51 of Chapter 5]

a. $\{(1,1,1,1),(1,-1,1,0),(1,1,-1,-1),(1,2,0,1)\}$.

b. $\{(1,2,2,1),(1,2,1,0),(0,0,1,1),(1,0,1,0)\}$.

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b. $\{(1,2,2,1),(1,2,1,0),(0,0,1,1),(1,0,1,0)\}$.

Solution.

a. Let $\boldsymbol{u}_1=(1,1,1,1)$, $\boldsymbol{u}_2=(1,-1,1,0)$, $\boldsymbol{u}_3=(1,1,-1,-1)$, $\boldsymbol{u}_4=(1,2,0,1)$. Then

$$\begin{aligned}
\boldsymbol{v}_1&=\boldsymbol{u}_1=(1,1,1,1),\\
\boldsymbol{v}_2&=\boldsymbol{u}_2-\frac{\boldsymbol{u}_2\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1=\left(\tfrac{3}{4},-\tfrac{5}{4},\tfrac{3}{4},-\tfrac{1}{4}\right),\\
\boldsymbol{v}_3&=\boldsymbol{u}_3-\frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1-\frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2=\left(\tfrac{14}{11},\tfrac{6}{11},-\tfrac{8}{11},-\tfrac{12}{11}\right),\\
\boldsymbol{v}_4&=\boldsymbol{u}_4-\frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1-\frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2-\frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_3}{\|\boldsymbol{v}_3\|^2}\boldsymbol{v}_3=\left(\tfrac{1}{10},-\tfrac{1}{10},-\tfrac{1}{5},\tfrac{1}{5}\right).
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Solution.

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\boldsymbol{v}_1 &= \boldsymbol{u}_1 = (1,1,1,1),\\
\boldsymbol{v}_2 &= \boldsymbol{u}_2 - \frac{\boldsymbol{u}_2\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 = \left(\tfrac{3}{4}, -\tfrac{5}{4}, \tfrac{3}{4}, -\tfrac{1}{4}\right),\\
\boldsymbol{v}_3 &= \boldsymbol{u}_3 - \frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \left(\tfrac{14}{11}, \tfrac{6}{11}, -\tfrac{8}{11}, -\tfrac{12}{11}\right),\\
\boldsymbol{v}_4 &= \boldsymbol{u}_4 - \frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 - \frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_3}{\|\boldsymbol{v}_3\|^2}\boldsymbol{v}_3 = \left(\tfrac{1}{10}, -\tfrac{1}{10}, -\tfrac{1}{5}, \tfrac{1}{5}\right).
\end{aligned}$$

Ans: $\left\{(1,1,1,1), \left(\tfrac{3}{4}, -\tfrac{5}{4}, \tfrac{3}{4}, -\tfrac{1}{4}\right), \left(\tfrac{14}{11}, \tfrac{6}{11}, -\tfrac{8}{11}, -\tfrac{12}{11}\right), \left(\tfrac{1}{10}, -\tfrac{1}{10}, -\tfrac{1}{5}, \tfrac{1}{5}\right)\right\}$

Question 3b: Solution

b. $\{(1,2,2,1),(1,2,1,0),(0,0,1,1),(1,0,1,0)\}$.

Let $\boldsymbol{u}_1 = (1,2,2,1)$, $\boldsymbol{u}_2 = (1,2,1,0)$, $\boldsymbol{u}_3 = (0,0,1,1)$, $\boldsymbol{u}_4 = (1,0,1,0)$. Then

$$\boldsymbol{v}_1 = \boldsymbol{u}_1 = (1,2,2,1),$$
$$\boldsymbol{v}_2 = \boldsymbol{u}_2 - \frac{\boldsymbol{u}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 = \left(\frac{3}{10}, \frac{3}{5}, -\frac{2}{5}, -\frac{7}{10}\right),$$
$$\boldsymbol{v}_3 = \boldsymbol{u}_3 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \boldsymbol{0} \quad \text{(reject)},$$
$$\boldsymbol{v}_3 = \boldsymbol{u}_4 - \frac{\boldsymbol{u}_4 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_4 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \left(\frac{8}{11}, -\frac{6}{11}, \frac{4}{11}, -\frac{4}{11}\right).$$

Ans: $\left\{(1,2,2,1), \left(\frac{3}{10}, \frac{3}{5}, -\frac{2}{5}, -\frac{7}{10}\right), \left(\frac{8}{11}, -\frac{6}{11}, \frac{4}{11}, -\frac{4}{11}\right)\right\}$

Question 3b: Solution

b. $\{(1,2,2,1),(1,2,1,0),(0,0,1,1),(1,0,1,0)\}$.

Let $\boldsymbol{u}_1 = (1,2,2,1)$, $\boldsymbol{u}_2 = (1,2,1,0)$, $\boldsymbol{u}_3 = (0,0,1,1)$, $\boldsymbol{u}_4 = (1,0,1,0)$. Then

$$\boldsymbol{v}_1 = \boldsymbol{u}_1 = (1,2,2,1),$$
$$\boldsymbol{v}_2 = \boldsymbol{u}_2 - \frac{\boldsymbol{u}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 = \left(\frac{3}{10}, \frac{3}{5}, -\frac{2}{5}, -\frac{7}{10}\right),$$
$$\boldsymbol{v}_3 = \boldsymbol{u}_3 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \boldsymbol{0} \quad \text{(reject)},$$
$$\boldsymbol{v}_3 = \boldsymbol{u}_4 - \frac{\boldsymbol{u}_4 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_4 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \left(\frac{8}{11}, -\frac{6}{11}, \frac{4}{11}, -\frac{4}{11}\right).$$

Ans: $\left\{(1,2,2,1), \left(\frac{3}{10}, \frac{3}{5}, -\frac{2}{5}, -\frac{7}{10}\right), \left(\frac{8}{11}, -\frac{6}{11}, \frac{4}{11}, -\frac{4}{11}\right)\right\}$

Question 3b: Solution

b. $\{(1,2,2,1),(1,2,1,0),(0,0,1,1),(1,0,1,0)\}$.

Let $\boldsymbol{u}_1 = (1,2,2,1)$, $\boldsymbol{u}_2 = (1,2,1,0)$, $\boldsymbol{u}_3 = (0,0,1,1)$, $\boldsymbol{u}_4 = (1,0,1,0)$. Then

$$\boldsymbol{v}_1 = \boldsymbol{u}_1 = (1,2,2,1),$$
$$\boldsymbol{v}_2 = \boldsymbol{u}_2 - \frac{\boldsymbol{u}_2 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 = \left(\frac{3}{10}, \frac{3}{5}, -\frac{2}{5}, -\frac{7}{10}\right),$$
$$\boldsymbol{v}_3 = \boldsymbol{u}_3 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_3 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \boldsymbol{0} \quad \text{(reject)},$$
$$\boldsymbol{v}_3 = \boldsymbol{u}_4 - \frac{\boldsymbol{u}_4 \cdot \boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_4 \cdot \boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \left(\frac{8}{11}, -\frac{6}{11}, \frac{4}{11}, -\frac{4}{11}\right).$$

Ans: $\left\{(1,2,2,1), \left(\frac{3}{10}, \frac{3}{5}, -\frac{2}{5}, -\frac{7}{10}\right), \left(\frac{8}{11}, -\frac{6}{11}, \frac{4}{11}, -\frac{4}{11}\right)\right\}$

Question 3b: Solution

b. $\{(1,2,2,1),(1,2,1,0),(0,0,1,1),(1,0,1,0)\}$.

Let $\boldsymbol{u}_1 = (1,2,2,1)$, $\boldsymbol{u}_2 = (1,2,1,0)$, $\boldsymbol{u}_3 = (0,0,1,1)$, $\boldsymbol{u}_4 = (1,0,1,0)$. Then

$$\begin{aligned}
\boldsymbol{v}_1 &= \boldsymbol{u}_1 = (1,2,2,1),\\
\boldsymbol{v}_2 &= \boldsymbol{u}_2 - \frac{\boldsymbol{u}_2\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 = \left(\frac{3}{10},\ \frac{3}{5},\ -\frac{2}{5},\ -\frac{7}{10}\right),\\
\boldsymbol{v}_3 &= \boldsymbol{u}_3 - \frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \boldsymbol{0} \quad \text{(reject)},\\
\boldsymbol{v}_3 &= \boldsymbol{u}_4 - \frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 - \frac{\boldsymbol{u}_4\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 = \left(\frac{8}{11},\ -\frac{6}{11},\ \frac{4}{11},\ -\frac{4}{11}\right).
\end{aligned}$$

Ans: $\left\{(1,2,2,1),\left(\frac{3}{10},\ \frac{3}{5},\ -\frac{2}{5},\ -\frac{7}{10}\right),\left(\frac{8}{11},\ -\frac{6}{11},\ \frac{4}{11},\ -\frac{4}{11}\right)\right\}$

Question 4

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Let $S = \{(1,1,0,0),(0,2,1,0),(0,6,0,1),(0,0,3,-1)\}$. Find an orthogonal basis B for $\mathbb{R}^4$ such that it has a subset C which is a basis for $\mathrm{span}(S)$.

[Pg 91 of Chapter 3, Gram-Schmidt, orthogonal projection]

Question 4

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Let $S = \{(1,1,0,0),(0,2,1,0),(0,6,0,1),(0,0,3,-1)\}$. Find an orthogonal basis B for $\mathbb{R}^4$ such that it has a subset C which is a basis for $\text{span}(S)$.

[Pg 91 of Chapter 3, Gram-Schmidt, orthogonal projection]

Proof.

Performing Gram-Schmidt process to S, we get

$$\boldsymbol{v}_1 = (1,1,0,0),\ \boldsymbol{v}_2 = (-1,1,1,0),\ \boldsymbol{v}_3 = (-1,1,-2,1).$$

So $C = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ is an orthogonal basis for $\text{span}(S)$.
Let $\boldsymbol{v} = (1,0,0,0)$. Then $\boldsymbol{v} \notin \text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$. The projection of $\boldsymbol{v}$ onto $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ is

$$\boldsymbol{v}_\text{p} = \frac{\boldsymbol{v}\cdot\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|^2}\boldsymbol{v}_1 + \frac{\boldsymbol{v}\cdot\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|^2}\boldsymbol{v}_2 + \frac{\boldsymbol{v}\cdot\boldsymbol{v}_3}{\|\boldsymbol{v}_3\|^2}\boldsymbol{v}_3 = (\tfrac{41}{42}, \tfrac{1}{42}, -\tfrac{1}{21}, -\tfrac{1}{7}).$$

Thus $\boldsymbol{v} - \boldsymbol{v}_\text{p} = (\frac{1}{42}, -\frac{1}{42}, \frac{1}{21}, \frac{1}{7})$ is orthogonal to $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$. Thus, $B = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v} - \boldsymbol{v}_\text{p}\}$ is an orthogonal set of nonzero vectors, and is hence linearly independent. Since $|B| = 4 = \dim(4)$, B is a basis for $\mathbb{R}^4$. □

Alternative Solution.

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 6 & 0 & 1 \\ 0 & 0 & 3 & -1 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $T := \{(1,1,0,0),(0,2,1,0),(0,0,-3,1)\}$ is a basis for $\operatorname{span}(S)$ and $T \cup \{(0,0,0,1)\}$ is a basis for $\mathbb{R}^4$.

Perform Gram-Schmidt process to $T \cup \{(0,0,0,1)\}$:

$\boldsymbol{v}_1 = (1,1,0,0),\ \boldsymbol{v}_2 = (-1,1,1,0),\ \boldsymbol{v}_3 = (-1,1,-2,1),\ \boldsymbol{v}_4 = (\frac{1}{7},-\frac{1}{7},\frac{2}{7},\frac{6}{7})$.

Then $B = \{\boldsymbol{v}_1,\boldsymbol{v}_2,\boldsymbol{v}_3,\boldsymbol{v}_4\}$ is an othogonal basis for $\operatorname{span}(T \cup \{(0,0,0,1)\}) = \mathbb{R}^4$, and $C = \{\boldsymbol{v}_1,\boldsymbol{v}_2,\boldsymbol{v}_3\}$ is a basis for $\operatorname{span}(T) = \operatorname{span}(S)$. □

Alternative Solution.

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 6 & 0 & 1 \\ 0 & 0 & 3 & -1 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $T := \{(1,1,0,0),(0,2,1,0),(0,0,-3,1)\}$ is a basis for $\operatorname{span}(S)$ and $T \cup \{(0,0,0,1)\}$ is a basis for $\mathbb{R}^4$.

Perform Gram-Schmidt process to $T \cup \{(0,0,0,1)\}$:

$$\boldsymbol{v}_1 = (1,1,0,0),\ \boldsymbol{v}_2 = (-1,1,1,0),\ \boldsymbol{v}_3 = (-1,1,-2,1),\ \boldsymbol{v}_4 = (\tfrac{1}{7},-\tfrac{1}{7},\tfrac{2}{7},\tfrac{6}{7}).$$

Then $B = \{\boldsymbol{v}_1,\boldsymbol{v}_2,\boldsymbol{v}_3,\boldsymbol{v}_4\}$ is an othogonal basis for $\operatorname{span}(T \cup \{(0,0,0,1)\}) = \mathbb{R}^4$, and $C = \{\boldsymbol{v}_1,\boldsymbol{v}_2,\boldsymbol{v}_3\}$ is a basis for $\operatorname{span}(T) = \operatorname{span}(S)$. □

Alternative Solution.

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 6 & 0 & 1 \\ 0 & 0 & 3 & -1 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $T := \{(1,1,0,0),(0,2,1,0),(0,0,-3,1)\}$ is a basis for $\operatorname{span}(S)$ and $T \cup \{(0,0,0,1)\}$ is a basis for $\mathbb{R}^4$.

Perform Gram-Schmidt process to $T \cup \{(0,0,0,1)\}$:

$$\boldsymbol{v}_1 = (1,1,0,0),\ \boldsymbol{v}_2 = (-1,1,1,0),\ \boldsymbol{v}_3 = (-1,1,-2,1),\ \boldsymbol{v}_4 = (\tfrac{1}{7},-\tfrac{1}{7},\tfrac{2}{7},\tfrac{6}{7}).$$

Then $B = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4\}$ is an othogonal basis for $\operatorname{span}(T \cup \{(0,0,0,1)\}) = \mathbb{R}^4$, and $C = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ is a basis for $\operatorname{span}(T) = \operatorname{span}(S)$. □

Next Lesson ...

We will explore **Sections 5.4 and 5.5**.
Please watch/review the video clips for these sections before then.